

Def: $X \sim \text{subG}(\sigma)$

$$\mathbb{E} \left(e^{\lambda(X - \mathbb{E}(X))} \right) \leq e^{\lambda^2 \sigma^2 / 2} \quad \lambda \in \mathbb{R}$$

$$\lambda > 0 \quad \mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda > 0} \left\{ \mathbb{E} \left(e^{\lambda(X - \mu)} \right) \cdot e^{-\lambda t} \right\}$$

$$\begin{aligned} & \leq \frac{e^{\lambda^2 \sigma^2 / 2}}{e^{\lambda^2 \sigma^2 / 2 - \lambda t}} \\ & \leq \inf_{\lambda > 0} e^{\lambda^2 \sigma^2 / 2 - \lambda t} \end{aligned}$$

$$\begin{aligned} \lambda^* &= t / \sigma^2 \\ &= e^{-t^2 / 2\sigma^2} \end{aligned}$$

subG

$$\mathbb{P}(X \geq t) \leq \mathbb{E}(X) / t$$

$$\mathbb{P}(e^{\lambda(X - \mu)} \geq e^{\lambda t}) \leq \mathbb{E}(e^{\lambda(X - \mu)}) \cdot e^{-\lambda t}$$

$$\mathbb{P}(X - \mu \geq t)$$

□

$$\mathbb{P}(|X - \mu| \geq t)$$

$$= \mathbb{P}(\underbrace{X - \mu \geq t}_A \text{ or } \underbrace{X - \mu \leq -t}_B)$$

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B) \quad \text{union bound}$$

$$\leq \mathbb{P}(X - \mu \geq t) + \mathbb{P}(X - \mu \leq -t)$$

$$\leq e^{-t^2/2\sigma^2}$$

$$\mathbb{P}(\underbrace{-X + \mu}_{X' - \mathbb{E}(X')} \geq t)$$

$$\mathbb{E}(e^{\lambda(X - \mu)}) \leq e^{\lambda^2 \sigma^2 / 2} \leq e^{-t^2/2\sigma^2}$$

$$\mathbb{E}(e^{\lambda(X' - \mu_{X'})}) = \mathbb{E}(e^{-\lambda(X - \mu)})$$

$$\leq e^{\lambda^2 \sigma^2 / 2}$$

$$\Rightarrow \mathbb{P}(|X - \mu| \geq t) \leq 2e^{-t^2/2\sigma^2}$$

□

$$X \in \{-1, 1\} \quad X \sim \text{Rad}$$

$$\mathbb{E}(X) = 0$$

$$\mathbb{E}(e^{\lambda X}) = \frac{1}{2} e^{\lambda} + \frac{1}{2} e^{-\lambda}$$

$$\lambda \in \mathbb{R} \quad e^{\lambda} = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = \frac{1}{2} \left[\sum_{n=0}^{\infty} \frac{\lambda^n}{n!} + \frac{(-\lambda)^n}{n!} \right]$$

$$= \sum_{n=0}^{\infty} \frac{\lambda^{2n}}{(2n)!}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{\lambda^{2n}}{(2n)!} \leq 1 + \sum_{n=1}^{\infty} \frac{\lambda^{2n}}{2^n n!}$$

$$(2n)! \geq n! 2^n$$

$$= \sum_{n=0}^{\infty} \frac{\left(\frac{\lambda^2}{2}\right)^n}{n!} = e^{\lambda^2/2}$$

$$\Rightarrow X \sim \text{Rad} \sim \text{Sub G} (\sigma=1)$$

Example 2

$$X \in [a, b] \sim \text{Sub G}$$

$$\mathbb{E} \left(e^{\lambda (X - \mathbb{E}(X))} \right)$$

$$X' \perp\!\!\!\perp X$$

$$X' \stackrel{d}{=} X$$

$$= \mathbb{E} \left(\underbrace{e^{\lambda (X - \mathbb{E}(X'))}}_f \right) \leq \mathbb{E}_X \mathbb{E}_{X'} e^{\lambda (X - X')}$$

Jensen's Ineq. f Convex

$$\mathbb{E} (f(X)) \geq f(\mathbb{E}(X))$$

$$E \sim \text{Rad}, \\ E \perp\!\!\!\perp X, X'$$

$$E(X - X') \stackrel{d}{=} X - X'$$

$$\Rightarrow \mathbb{E} \left(e^{\lambda (X - \mathbb{E}(X))} \right) \leq \mathbb{E}_X \mathbb{E}_{X'} \mathbb{E}_\epsilon e^{\lambda \epsilon (X - X')}$$

$$\begin{aligned} \text{Jensen's} \quad \mathbb{E}_{X,Y} (g(X,Y)) &= \mathbb{E}_X \mathbb{E}_{Y|X} (g(X,Y) | X) \\ &= \mathbb{E}_{X,X'} \mathbb{E}_{\epsilon|X,X'} e^{\lambda \epsilon (X - X')} \\ &\leq \mathbb{E}_{X,X'} \underbrace{e^{\frac{\lambda^2 (X - X')^2}{2}}}_{\leq e^{\frac{\lambda^2 (X - X')^2}{2}}} \\ &\leq \mathbb{E}_{X,X'} e^{\frac{\lambda^2 (X - X')^2}{2}} \\ &\leq e^{\frac{\lambda^2 (b-a)^2}{2}} \end{aligned}$$

$$\sigma^2 = (b-a)^2 \Rightarrow \sigma = b-a$$

Exercise 2.4

$$\sigma = (b-a)/2$$

□

Hoeffding's Ineq.

$X_1, \dots, X_n$ indep sub $G_i(\sigma_i)$

$$\mathbb{P} \left(\sum X_i - \mu_i \geq t \right) \leq e^{-\frac{t^2}{2 \sum \sigma_i^2}}$$

$X_1, \dots, X_n$ indep & sub $G_i(\sigma_i)$

$$X_1 + \dots + X_n \sim \text{SubG}(\sqrt{\sigma_1^2 + \dots + \sigma_n^2})$$

$$\mathbb{E} \left(e^{\lambda (X_1 + X_2 - \mathbb{E}(X_1) - \mathbb{E}(X_2))} \right)$$

$$= \mathbb{E} \left(e^{\lambda (X_1 - \mathbb{E}(X_1))} \cdot e^{\lambda (X_2 - \mathbb{E}(X_2))} \right)$$

$$= \mathbb{E} \left(e^{\lambda (X_1 - \mathbb{E}(X_1))} \right) \cdot \mathbb{E} \left(e^{\lambda (X_2 - \mathbb{E}(X_2))} \right)$$

$$\begin{aligned} & \underbrace{\leq e^{\lambda^2 \sigma_1^2 / 2}}_{\leq e^{\lambda^2 \sigma_1^2 / 2}} \cdot \underbrace{\leq e^{\lambda^2 \sigma_2^2 / 2}}_{\leq e^{\lambda^2 \sigma_2^2 / 2}} = e^{\lambda^2 (\sigma_1^2 + \sigma_2^2) / 2} \\ & \sigma^2 = \sigma_1^2 + \sigma_2^2 \Rightarrow \sigma = \sqrt{\sigma_1^2 + \sigma_2^2} \end{aligned}$$

$$Y = \sum X_i \sim \text{SubG}(\sqrt{\sigma_1^2 + \dots + \sigma_n^2})$$

$$\mathbb{P} \left(Y - \mu_Y \geq t \right) \leq \exp \left\{ - \frac{t^2}{2 \sum \sigma_i^2} \right\}$$

$$\mathbb{P} \left(\sum X_i - \mu_i \geq t \right)$$

$$\mathbb{P}\left(\frac{1}{n} \sum x_i - \mu \geq \underbrace{\frac{t}{n}}_{t'}\right) \leq \exp\left\{-\frac{t^2}{2n\sigma^2}\right\}$$

$$\begin{aligned} \mathbb{P}\left(\frac{1}{n} \sum x_i - \mu \geq t'\right) &\leq \exp\left\{-\frac{n^2 t'^2}{2n\sigma^2}\right\} \\ &= \exp\left\{-\frac{nt'^2}{2\sigma^2}\right\} \end{aligned}$$

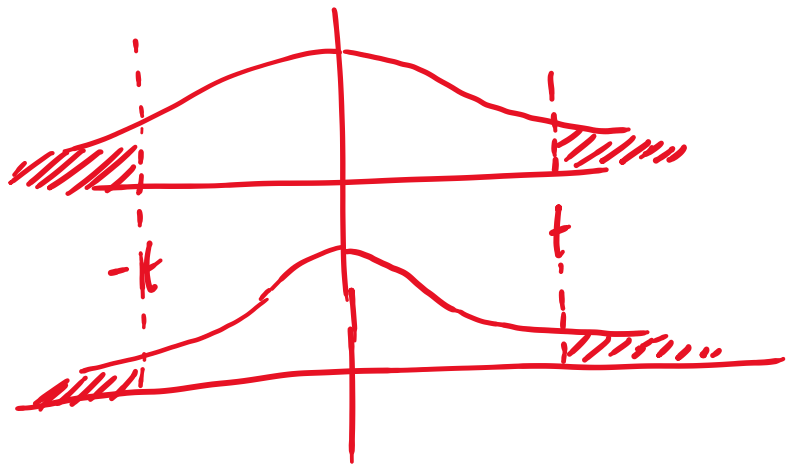
$$\mathbb{P}\left(\left|\underbrace{\frac{1}{n} \sum x_i}_{\text{empirical mean}} - \underbrace{\mu}_{\text{population mean}}\right| \geq \epsilon\right) \leq 2e^{-\frac{n\epsilon^2}{2\sigma^2}}$$

□

Example : $X \sim \text{subG}(\sigma)$, $\mathbb{E}(X) = 0$.

$$\exists c > 0. \quad \mathbb{P}(|X| \geq t) \leq c \cdot \mathbb{P}(|Z| \geq t)$$

$$Z \sim \mathcal{N}(0, \underbrace{\tau^2}_{2\sigma^2})$$



$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$P(Z \geq t) \geq \left(\frac{1}{t} - \frac{1}{t^3}\right) \phi(t)$$

$$\phi'(z) = -\frac{1}{\sqrt{2\pi}} z e^{-z^2/2} = -z \phi(z)$$

$$\phi'(z) + z \phi(z) = 0$$

$$P(Z \geq t) = \int_t^{\infty} \phi(z) dz$$

$$= \int_t^{\infty} \underbrace{-\frac{1}{z} \phi'(z)}_{dv} dz$$

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

$$\begin{cases} v = \phi(z) \\ u = -1/z \end{cases}$$

$$\frac{-\phi(z)}{z} \Big|_t^{\infty} - \int_t^{\infty} \frac{\phi(z)}{z^2} dz$$

$$= \frac{\phi(t)}{t} + \int_t^{\infty} \frac{1}{z^3} \phi'(z) dz$$

$$= \phi(t)/t - \phi(t)/t^3 - \int_t^{\infty} \frac{3}{z^4} \phi(z) dz$$